Alaap Joshi

alaapjoshi@gmail.com | (908) 952-5450 | alaapjoshi.github.io | linkedin.com/in/alaapjoshi

Education

Rutgers University–New Brunswick — New Brunswick, NJ
B.S. in Computer Science and Cognitive Science — Expected May 2029

Relevant Coursework: Intro to Computer Science, Data Structures, Data Management for Data Science, Computer Architecture, Discrete Structures I, Linear Algebra, Calculus I–II, Physics I–II, The Structure of Information,
Achievements: Top **10** Data Science Competition @ MIT; **15** hackathon awards across **22** competitions

Professional Experience

Robert Wood Johnson Medical School | Undergraduate Research Assistant — **Jul 2026 – Present**
- Analyzed **250+ fluorescence microscopy images** with Python, scikit-image, and Cellpose for cell segmentation.
- Automated thresholding pipelines that cut manual image-analysis time by **23%** and quantified cell area and MFI.

Ipser Labs | Software Engineering Intern – Remote — **Jun 2026 – Aug 2026**
- Built React Native scanning and validation workflows that digitized **1,000+ land-ownership and debt records**.
- Optimized React Native assets and dependencies to reduce application bundle size by **11%**, improving load performance, deployment efficiency, and consistency across customer environments.

theCoderSchool | Tutor and Mentor – Montgomery, NJ — **May 2024 – Aug 2025**
- Taught **60+ students** ages 6–18 Python, Java, web development, and algorithms through hands-on projects.
- Delivered **200+ private and group sessions** using project-based lessons, debugging, and coding challenges.

Leadership

Rutgers Organization of Cloud Computing (ROCC) | Board Member — **Jan 2026 – Present**
- Develop AWS and Azure curriculum and plan technical workshops covering cloud architecture and deployment.
- Build hands-on cloud projects that give members practice with networking, storage, infrastructure, troubleshooting, deployment, and application management across AWS and Azure environments through guided workshops.

Hillsborough High School Computer Science Club | President — **2023 – 2024**
- Led programming workshops and project teams while mentoring students in software development and hackathons.
- Organized coding activities and guided teams through project planning, implementation, testing, debugging, presentation, and competition preparation, helping students turn ideas into polished software demos.

FIRST Robotics Team 75 | App Development Lead — **2022 – 2024**
- Led React Native/Firebase scouting-app development, centralizing match observations for competition strategy.
- Coordinated cross-platform testing and data-entry workflows so scouts could capture consistent match data across phones, sync results to Firebase, and review centralized observations for competition strategy.

Projects

Intellibin | Arduino, YOLOv5, PyTorch, 3D Printing — **Hack MHS III – 1st Place**
- Built a YOLOv5 smart bin that classified **23 waste types** and controlled Arduino-driven sorting flaps in real time.
- Integrated PyTorch inference with a 3D-printed enclosure and motorized hardware to route detected waste into separate compartments by predicted material category, connecting model output to real-time sorting.

ZardonVision | SwiftUI, UIKit, Flask, PyTorch — **MontyHacks VI – 2nd Place**
- Built a voice-controlled accessibility app using OCR and computer vision to generate spoken image descriptions.
- Connected a SwiftUI/UIKit client to a Flask and PyTorch backend for image processing and accessibility feedback.

75watch | React Native, Expo, Firebase — **FIRST Robotics Competition**
- Built a React Native scouting app that captured match observations and synced results to Firebase in real time.
- Designed a cross-platform data-entry workflow so scouts could record matches quickly, sync observations to Firebase, and review centralized results after events for scouting analysis and competition strategy.

Technical Skills

Languages: Python, Java, JavaScript, TypeScript, C++, SQL, HTML/CSS, Swift, Bash, LaTeX
Web/Mobile: React, React Native, Expo, Flask, FastAPI, Firebase, PostgreSQL, Svelte
ML/Data: PyTorch, NumPy, pandas, scikit-image, Cellpose, OpenCV, Jupyter Notebook
Tools/Platforms: Git, Docker, AWS, Arduino, Xcode, Android Studio, Linux, Wireshark
Interests: Inter Milan, Rap/R&B, Tennis, Streetwear, Golf, Sneaker Collecting, Tabla, Concerts, Chess,